

RESEARCH PAPER

A Transformer-Based Multimodal Cross-Attention Framework with Relational Graph Attention for Pill–Prescription Matching

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Abstract: Matching physical pills with drug names on a prescription is a safety-critical task for preventing medication errors. The pioneering PIMA (Pill-prescription MAtching) framework addressed this problem by aligning convolutional features (ResNet50) with textual embeddings (BERT) using a direction-agnostic GraphSAGE operator and a margin-based contrastive loss. While effective on controlled datasets, PIMA’s real-world deployment is hindered by its reliance on manual ground-truth annotations, sensitivity to background clutter, and a simplistic cosine similarity fusion that lacks selective attention. In this paper, we propose a redesigned, fully multimodal framework featuring five coordinated improvements: (i) two interchangeable visual encoders for the pill-recognition branch—a Vision Transformer (ViT-B/16) applied to background-removed crops, and an end-to-end Faster R-CNN/RoI-Align detector that learns features directly from raw images; (ii) a lightweight all-MiniLM-L6-v2 sentence encoder for efficient pill-name representation; (iii) an integrated PP-OCRv3 module that automates text and bounding box extraction, eliminating manual dependency; (iv) a Relational Graph Attention Network (R-GAT) with a pseudo-classifier gate to model directional spatial logic (e.g., left-to-right drug/quantity relations) and filter irrelevant non-drug text; and (v) a multimodal cross-attention head optimized via an InfoNCE objective for more stable and dense gradient convergence. Experiments on the real-world VAIPE dataset show that both variants substantially outperform a reproduced graph baseline: the ViT-B/16 variant raises the Top-1 matching accuracy from 49.89% to 82.46% and the end-to-end Faster R-CNN variant to 83.47%—absolute gains of 32.57 and 33.58 percentage points (pp), respectively. The two variants trade off accuracy against preprocessing: the Faster R-CNN variant attains the best accuracy while operating directly on raw images, whereas the ViT-B/16 variant requires a separate background-removal step. Our system also remains robust to optical-character-recognition (OCR) noise, showing minimal accuracy degradation when switching from ground-truth to automated text extraction. These results confirm that a Transformer-based cross-attention architecture with directional spatial awareness provides an effective and deployable solution for automated pill–prescription verification.

Keywords:

Pill–prescription matching, Multimodal learning, Vision Transformer, Relational Graph Attention, Cross-attention, InfoNCE, Medication safety

1 Introduction

Medication errors represent a persistent and global threat to patient safety [1]. According to the World Health Organization (WHO), avoidable harm in healthcare is frequently attributable to the administration of incorrect drugs, wrong dosages, or non-prescribed medications [2]. Statistics indicate that several thousand deaths annually—approximately 6,000 to 8,000 in certain reports—are linked directly to these mistakes [2, 3]. Recognizing the severity of this issue, the WHO designated “Medication Without Harm” as the



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theme for World Patient Safety Day 2022 [2]. Automated verification systems that can validate medications from ordinary mobile photographs against written prescriptions therefore hold immense clinical value [1]. Consequently, pill image recognition has transitioned from a niche task to an active research area spanning computer vision, optical character recognition (OCR), and clinical informatics [1].

While deep learning models have revolutionized isolated pill classification, real-world verification requires more than identifying a single pill. A recent comprehensive survey [2] highlights that the most realistic and challenging formulation is not classification in a controlled setting, but the joint reasoning required to connect a photograph of multiple pills to the structured-but-noisy text of a prescription. This task, formalized as pill–prescription matching, requires a model to bridge the semantic gap between visual pill morphology and textual drug nomenclature [3]. Unlike general text-image matching, drug names often lack inherent semantic meaning, are lengthy, and are embedded within a layout filled with irrelevant information such as dosages and diagnoses [3, 4]. This line of work originates from a sustained research effort on Vietnamese real-world pill data: contextual single-pill recognition aided by a medical knowledge graph (PIKA) [5], class-incremental pill classification [6], and graph-assisted multi-pill detection on the open VAIPE dataset [7]. Within this effort, the PIMA (Pill-prescription MAtching) framework [3] was the first to define and address the pill–prescription matching task using a graph-based representation of prescriptions, and it was later extended to a zero-shot setting (Zero-PIMA) [4].

PIMA [3] modeled prescriptions as graphs whose nodes represent text boxes and whose edges capture spatial proximity, using a GraphSAGE operator to identify drug names. Its architecture, however, has four bottlenecks that limit real-world deployment. It relies on a convolutional backbone (ResNet50) [8] that is sensitive to background clutter and struggles to capture fine-grained morphological cues. It also assumes access to manual ground-truth annotations for text and bounding boxes, which makes it non-functional for end-to-end use without human intervention. In addition, it aggregates spatial context with a direction-agnostic operator that cannot distinguish directional layout logic, such as a dosage quantity typically appearing to the right of a drug name. Lastly, it fuses the two modalities with a simple cosine similarity that lacks selective attention, preventing the model from focusing on the most relevant textual evidence for a given pill’s appearance.

In this work, we argue that a Transformer-based multimodal design can jointly overcome these limitations. We propose a redesigned framework featuring: (i) a Vision Transformer (ViT-B/16) for global feature extraction and a Faster R-CNN/RoI-Align variant for end-to-end processing of raw images [9, 10]; (ii) a compact all-MiniLM-L6-v2 sentence encoder for efficient textual representation; (iii) an integrated PP-OCRv3 reader to automate the entire text extraction pipeline [11]; (iv) a Relational Graph Attention Network (R-GAT) with a pseudo-classifier gate to model directional spatial relations and filter non-drug text; and (v) a multimodal cross-attention head optimized via InfoNCE for stable gradient convergence. Unlike previous pair-wise margin losses, InfoNCE contrasts positive pairs against all negative samples in a batch, providing a much denser learning signal.

The main contributions of this paper are summarized as follows: we identify and analyze the architectural bottlenecks of the original PIMA framework and propose a coherent set of five enhancements to address them; we design a fully multimodal architecture coupling Transformer-based encoders with an R-GAT spatial encoder and a cross-attention fusion head; we integrate automated OCR and demonstrate that our framework, evaluated on the real-world VAIPE dataset [7], raises Top-1 matching accuracy from 49.89% (for the reproduced graph baseline) to 82.46% (ViT-B/16) and 83.47% (Faster R-CNN); and we provide an ablation study confirming that our system remains robust to OCR noise, maintaining high performance even when processing raw, unprocessed photographs, thus providing a deployable solution for automated pill verification.

The remainder of this paper is organised as follows. Section 2 reviews related work on pill recognition,

object detection, graph neural networks, and multimodal contrastive learning. Section 3 details the proposed framework and its differences from the baseline. Section 4 describes the dataset, implementation, and evaluation protocol. Section 5 reports and discusses the results, including training curves and an ablation study. Section 6 concludes the paper and outlines future work.

2 Related Work

Our work builds on four threads of prior research: pill image recognition, object detection and visual backbones, graph neural networks (GNNs) for structured document understanding, and multimodal contrastive learning. We follow the taxonomy established in the recent comprehensive survey of pill image recognition [2].

2.1 Pill Image Recognition

Early pill-identification systems relied on hand-crafted descriptors of the three canonical pill attributes—shape, colour, and imprint [2]. Shape was commonly captured with edge detectors such as Canny [12] and with moment invariants [13]; imprint and texture were described with local features such as SIFT [14] and Local Binary Patterns (LBP) [15]. While interpretable, such pipelines are inherently brittle: their performance degrades sharply when the imprint is worn, the lighting changes, or the background is cluttered, requiring careful manual tuning for each new dataset [2].

The subsequent shift to deep learning replaced these manual descriptors with features learned end-to-end. Convolutional networks such as VGG [16] and ResNet [8] became standard backbones for pill classification and retrieval, and underpin dedicated pill benchmarks and inspection models such as the ePillID benchmark [17] and deep pill-detection systems [18]. To cope with the fine-grained, low-inter-class-variance nature of medications, multi-stream variants that fuse several views or modalities have been proposed, including class-incremental multi-stream fusion [6] and metric-based recognition that combines textual and visual cues [19]. The most recent systems push this direction further toward deployable, real-world pipelines: clinical multi-platform studies of code-free deep learning [20], mobile medication identification [21], and real-time inspection frameworks [22]. Closely related to our setting, a multimodal framework that fuses RGB, contour, texture, and text cues through an adaptive lightweight-attention mechanism has been shown to achieve accurate real-time pill recognition on resource-constrained edge devices [23], underscoring the value of attention-based multimodal fusion that we also adopt in this work. In parallel, the field has begun to adopt Transformer-based recognition [2]. This trend motivates our adoption of a Vision Transformer (ViT) backbone, which uses self-attention to capture global morphological cues that traditional convolutional neural networks (CNNs) often struggle to resolve in noisy or unconstrained environments.

2.2 From Single-Pill Recognition to Pill–Prescription Matching

Recognising an isolated pill is insufficient for verifying a prescription: the system must also decide *which* prescribed drug each detected pill corresponds to. This problem has been driven largely by a single, sustained line of research on real-world Vietnamese pill data. The group first introduced contextual single-pill recognition with a medical knowledge graph (PIKA) [5], then addressed catastrophic forgetting through a multi-stream class-incremental learning framework (CG-IMIF) [6], and tackled multi-pill localisation with a graph-assisted, explainable detection framework that also released the large-scale, open VAIPE dataset used in this study [7].

Within this effort, Nguyen et al. defined the pill–prescription matching task together with the PIMA (Pill-prescription MAtching) framework [3]. PIMA builds a graph over the text boxes of a prescription, propagates information with GraphSAGE to emphasise boxes containing pill names, and matches the highlighted text embeddings against convolutional pill features with a contrastive loss, supported by an auxiliary classification loss. The framework was subsequently extended to recognise pills unseen during training in a zero-shot

setting (Zero-PIMA), where a graph convolutional network and metadata-enriched name embeddings improve matching for novel drugs [4]. PIMA is the baseline that the present paper improves upon: we retain its problem formulation and its use of a text-box graph, but we replace its visual backbone, its graph operator, its language encoder, and its fusion mechanism, and we remove its dependence on ground-truth annotations.

2.3 Object Detection and Visual Backbones

Because pills must be localised within a photograph before they can be matched, object detection is central to an end-to-end system. Two-stage detectors such as Faster R-CNN [9], strengthened with feature pyramid networks [10] and RoI-Align region features [24], offer accurate localisation, while one-stage detectors such as [25], SSD [26], and [27] trade some accuracy for speed. Segmentation networks such as U-Net [28] and Mask R-CNN [24] are frequently used to isolate pills from complex backgrounds [2]. In our end-to-end variant we adopt a Faster R-CNN/ResNet50-FPN detector and use RoI-Align to extract a feature vector for each pill directly from the raw image, avoiding a separate background-removal stage.

2.4 Text Extraction from Prescriptions

Reading the drug names from a prescription requires optical character recognition. Classical engines such as Tesseract [29] perform well on clean, printed text but struggle with the noisy, heterogeneous layouts of real prescriptions. Modern OCR systems based on convolutional–Transformer hybrids, in particular PP-OCRv3 [11], combine text detection and recognition and are far more robust under such conditions [2]. We embed PP-OCRv3 in our pipeline so that text content and bounding boxes are produced automatically, replacing the ground-truth annotations assumed by the baseline.

2.5 Graph Neural Networks and Multimodal Contrastive Learning

Graph neural networks are a natural tool for the spatial layout of a prescription, where the relevant signal is the relative position of text boxes. The baseline uses GraphSAGE, which aggregates neighbours without regard to their direction. Relational graph attention, by contrast, learns separate attention weights for typed (directional) edges, allowing the model to encode that a quantity usually appears to the right of a drug name or that a diagnosis appears above the medication list. For the cross-modal step, contrastive objectives that pull matched image–text pairs together while pushing apart mismatched pairs have become the de-facto standard; the InfoNCE loss in particular optimises a softmax over all pairs in a batch and provides a stronger learning signal than a fixed cosine threshold. Our framework combines a relational graph attention encoder with a cross-attention fusion module and an InfoNCE objective, an instantiation of these ideas tailored to pill–prescription matching.

3 Proposed Method

3.1 Overview and Problem Formulation

Given a prescription image P and an associated image I containing one or more pills, the goal of pill–prescription matching is to assign to every detected pill v the drug name on the prescription that it corresponds to, or to report that no match exists. Let the prescription contain M text boxes with textual content $\{t_1, \dots, t_M\}$ and spatial coordinates $\{b_1, \dots, b_M\}$, and let the pill image yield N pill regions $\{x_1, \dots, x_N\}$. The model must produce, for each pill x_i , a probability distribution over the candidate drug-name boxes.

The proposed framework, shown in Fig. 1, comprises three encoders and a fusion head. A visual encoder maps each pill region to a feature vector; a text encoder maps each prescription text box to a sentence embedding; a relational graph encoder refines these embeddings using the directional spatial layout of the prescription and predicts, through a pseudo-classifier, which boxes are likely to contain pill names; and a multimodal cross-attention head fuses the visual and textual representations and is trained with an InfoNCE

contrastive objective. Unlike the baseline PIMA [3], the text content and bounding boxes are not assumed to be given: they are produced automatically by an integrated PP-OCRv3 reader [11].

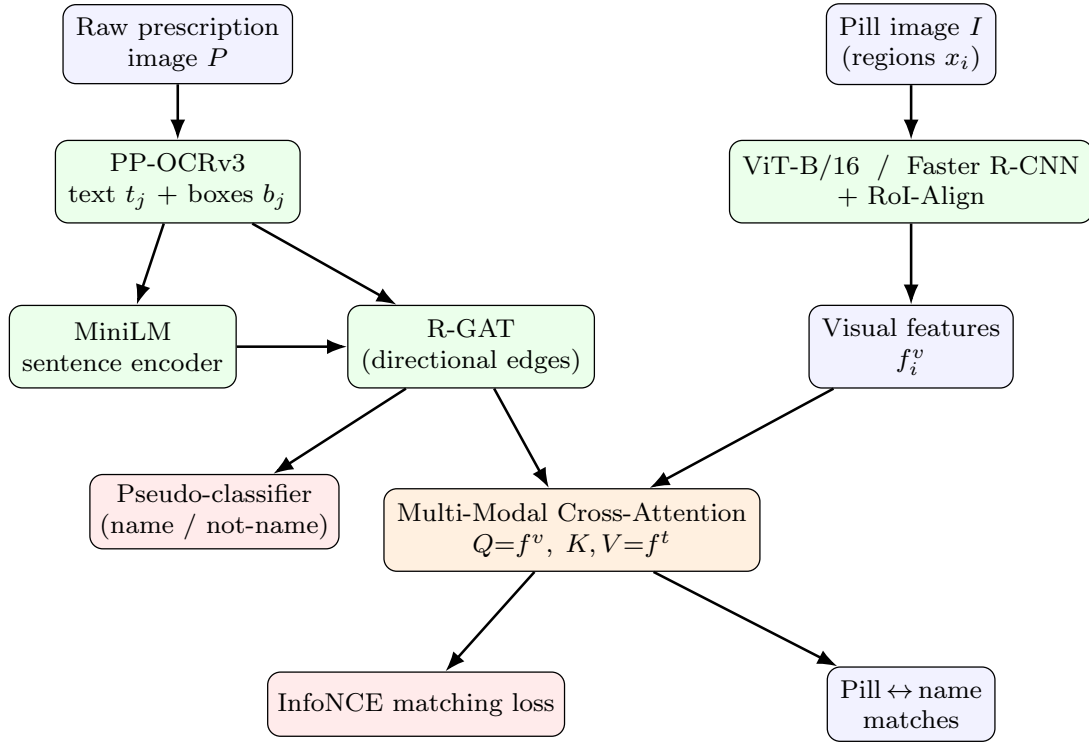


Figure 1: Overview of the proposed Transformer-based multimodal framework. Green blocks are learnable encoders, the orange block is the cross-attention fusion head, and red blocks are training objectives. The PP-OCRv3 reader removes the dependence on ground-truth text and bounding boxes assumed by the baseline.

3.2 Visual Encoder

The original PIMA framework extracts pill features with a ResNet50 convolutional backbone. We replace this with two interchangeable Transformer-aware alternatives.

ViT-B/16 (clean-crop variant). Each pill is first separated from its background (e.g. with a matting tool) and the clean crop is fed to a Vision Transformer ViT-B/16. By splitting the image into 16×16 patches and modelling them with self-attention, the ViT captures global morphological cues—outline, colour distribution, and imprint—without being distracted by background texture.

The core operation driving this architecture is the Scaled Dot-Product Attention:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V \quad (1)$$

This allows the model to establish long-range dependencies across the entire pill image, ensuring that fine-grained details like imprints are not lost in background noise (which is a common limitation of the ResNet50 backbone). Finally, the class token produces a d -dimensional pill embedding f_i^v .

Faster R-CNN (end-to-end variant). To avoid a separate background-removal step, the second variant feeds the raw image, together with pill bounding boxes, to a Faster R-CNN with a ResNet50-FPN backbone [9, 10]. RoI-Align [24] extracts a fixed-size feature map for each pill region, which is pooled into the

embedding f_i^v . This variant is fully end-to-end and therefore better suited to deployment on unprocessed photographs, at the cost of having to learn to suppress background noise.

3.3 Language Encoder and OCR

Drug names are extracted from the prescription by PP-OCRv3 [11], which jointly detects text regions and recognises their content, yielding for each box j a string t_j and coordinates b_j . Each string is embedded by the compact all-MiniLM-L6-v2 sentence encoder, producing a textual feature $f_j^t \in \mathbb{R}^d$. Compared with the heavier BERT encoder used previously, MiniLM provides sentence embeddings at a fraction of the computational cost while preserving semantic quality, which is important because a prescription contains many boxes that must each be encoded.

3.4 Relational Graph Attention Encoder

The prescription is represented as a graph $G = (\mathcal{V}, \mathcal{E})$ whose nodes are the text boxes and whose edges encode *directional* spatial relations (left, right, above, below) derived from the boxes’ coordinates. Whereas the direction-agnostic GraphSAGE aggregates neighbours identically regardless of direction, we use a Relational Graph Attention (R-GAT) operator. This explicitly models the directional spatial logic of a prescription (e.g., drug name $\xrightarrow{\text{right}}$ quantity) by learning a separate transformation and attention coefficient per relation type r :

$$h_j^{(l+1)} = \sigma \left(\sum_{r \in \mathcal{R}} \sum_{k \in \mathcal{N}_r(j)} \alpha_{rjk} W_r h_k^{(l)} \right), \quad (2)$$

where $\mathcal{N}_r(j)$ is the set of neighbours of node j connected by relation r , and W_r is a relation-specific weight matrix. By using these distinct matrices for each direction, the model effectively reconstructs the logical structure of the document, which is critical for identifying which text boxes actually belong to a specific medication entry. The attention weight α_{rjk} is obtained by a softmax over a learnable compatibility score between $h_j^{(l)}$ and $h_k^{(l)}$. The node features are initialised with the MiniLM embeddings, $h_j^{(0)} = f_j^t$.

A *pseudo-classifier* gate is attached to the final node representations to filter out non-drug information like dates or diagnoses. Mathematically, it acts as a sigmoid-based attention gate:

$$p_j = \sigma(W_{\text{gate}} h_j^{(L)} + b_{\text{gate}}) \quad (3)$$

The model predicts, for each box, the probability p_j that it contains a pill name, and is trained with a binary cross-entropy loss $\mathcal{L}_{\text{cls}}$. This p_j value serves as a weight that down-regulates irrelevant nodes before they reach the fusion stage, preventing the model from being confused by “noise” text that often exists in real-world prescriptions.

3.5 Cross-Attention Fusion and InfoNCE Objective

Rather than comparing the two modalities with a fixed cosine similarity, we let each pill actively query the prescription. In the multimodal cross-attention head, the visual features f_v act as queries and the graph-refined textual features h act as keys and values:

$$\text{CrossAttn}(f_v, h) = \text{softmax} \left(\frac{(f_v W_Q)(h W_K)^\top}{\sqrt{d}} \right) (h W_V). \quad (4)$$

This mechanism enables active querying, where the pill’s appearance dynamically weights different parts of the prescription text to find the optimal match. Consequently, each pill can retrieve the textual evidence most consistent with its visual features, offering a more expressive operation than a simple dot product.

Let z_i^v and z_i^t denote the ℓ_2 -normalised projections of the matched pill and pill-name in a training batch of size B . To improve convergence stability, the framework replaces the traditional margin-based contrastive

loss of the baseline. The model is optimised with a symmetric InfoNCE loss with temperature τ :

$$\mathcal{L}_{\text{match}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(z_i^v \cdot z_i^t / \tau)}{\sum_{k=1}^B \exp(z_i^v \cdot z_k^t / \tau)}. \quad (5)$$

Unlike the margin-based approach, Eq. (5) contrasts each positive pair against all negative samples in the batch simultaneously. This provides a denser and more stable gradient signal, which contributes to the higher accuracy (from 49.89% to over 83%) compared with the baseline.

The overall training objective combines the matching and pseudo-classification terms,

$$\mathcal{L} = \mathcal{L}_{\text{match}} + \lambda \mathcal{L}_{\text{cls}}, \quad (6)$$

with λ balancing the two. The complete training procedure is summarised in Algorithm 1.

Algorithm 1 Training of the proposed framework

Input: prescription images $\{P\}$, pill images $\{I\}$, match labels.

For each mini-batch **do**:

1. $\{t_j, b_j\} \leftarrow \text{PP-OCRv3}(P)$ (text + bounding boxes)
2. $f_j^t \leftarrow \text{MiniLM}(t_j)$; build the directional graph G from $\{b_j\}$
3. $h_j \leftarrow \text{R-GAT}(G, f_j^t)$ via Eq. (2); $p_j \leftarrow \text{pseudo-classifier}(h_j)$
4. $f_i^v \leftarrow \text{ViT} / \text{Faster R-CNN}(I)$ (pill features)
5. $c_i \leftarrow \text{CrossAttn}(f_i^v, h)$ via Eq. (4)
6. compute $\mathcal{L}_{\text{match}}$ (Eq. (5)) and $\mathcal{L}_{\text{cls}}$; update parameters by minimising Eq. (6)

end for

3.6 Differences from the Baseline

Table 1 summarises, component by component, how the proposed framework differs from the original PIMA [3]. Every stage of the pipeline—vision, language, spatial reasoning, and fusion—is upgraded, and the dependence on manual annotation is removed.

Table 1: Component-wise comparison between the baseline PIMA [3] and the proposed framework.

Component	Baseline PIMA	Proposed framework
Visual encoder	ResNet50 (CNN) on cropped pills	ViT-B/16 on clean crops; or Faster R-CNN + RoI-Align end-to-end
Text input	Ground-truth text & boxes	Automatic PP-OCRv3 (text + boxes)
Language encoder	BERT	all-MiniLM-L6-v2
Spatial reasoning	GraphSAGE (direction-agnostic)	R-GAT (directional) + pseudo-classifier gate
Fusion	Cosine similarity	Multimodal cross-attention
Objective	Margin contrastive + NLL	InfoNCE + BCE (Eq. (6))

4 Experiments

4.1 Dataset

We evaluate our proposed framework on the real-world VAIP dataset [7], which presents significant practical challenges such as cluttered backgrounds, varied lighting, and noisy layouts containing both printed

and handwritten text. To ensure a rigorous and unbiased evaluation, the data is partitioned into two primary, non-overlapping subsets: a development pool for model optimization and a completely independent, hold-out test set for final evaluation.

The development pool comprises 10,675 image files, consisting of 1,173 raw prescription photographs and 9,502 pill images. This pool is split with an 8:2 ratio into a training and a validation set. The training set contains 8,540 images (938 prescriptions and 7,602 pill images) and is used to optimize the learnable parameters of the visual encoders, the R-GAT spatial encoder, and the cross-attention fusion head. The validation set contains the remaining 2,135 images (235 prescriptions and 1,900 pill images) and is used exclusively for hyperparameter tuning and the early-stopping criterion to prevent over-fitting.

The hold-out test set is a separate set of 3,219 images (358 prescriptions and 2,861 pill images) that remains unseen during all stages of training and validation; Table 2 summarizes this partition. Because a single photograph typically shows several pills, the test set yields more individual pills than photographs at evaluation: it provides 7,058 individual pill instances for the end-to-end Faster R-CNN variant and 3,627 instances for the ViT variant, which discards pills that cannot be cleanly segmented. These per-pill instances form the basis for the Top-1 matching accuracy reported in Section 5.

Table 2: Statistics of the VAIPE subset and the train/validation/test partition.

Subset	Prescriptions	Pill images	Total images
Training	938	7,602	8,540
Validation	235	1,900	2,135
Test (hold-out)	358	2,861	3,219
Total	1,531	12,363	13,894

4.2 Implementation Details

The visual encoders utilized in our framework are interchangeable: a ViT-B/16 (clean-crop variant) and a Faster R-CNN with a ResNet50-FPN backbone (end-to-end variant). Text extraction is automated using PP-OCRv3, and the textual content is embedded via the lightweight all-MiniLM-L6-v2 model. The spatial graph is processed through a Relational Graph Attention Network (R-GAT) whose final node features feed both a pseudo-classifier gate and the cross-attention head.

The training process was implemented in PyTorch and optimized using the AdamW optimizer with an initial learning rate of 5×10^{-5} and a weight decay of 10^{-4} . The learning rate was decayed following a Cosine Annealing schedule ($T_{max} = 80$). To handle the substantial computational requirements, models were trained under distributed data parallelism (torchrun) on a hardware setup utilizing 4 GPUs with a combined total of 64GB VRAM.

The base batch size was set to 2 per GPU; however, the end-to-end Faster R-CNN variant additionally utilized gradient accumulation over 4 steps to emulate a larger effective batch size and prevent memory bottlenecks. Furthermore, gradient clipping (max norm = 1.0) was applied to stabilize training. Early stopping was strictly enforced using a patience of 10 epochs based on the validation metric, which explains why the two backbones converged at different stages (epoch 45 for ViT, epoch 17 for Faster R-CNN).

4.3 Evaluation Protocol

Following the baseline, our primary metric is the *Top-1 matching accuracy*: the fraction of evaluated pills whose highest-scoring candidate strictly matches the correct drug name. We additionally monitor the training and validation accuracy and loss recorded at each epoch to characterize convergence and over-fitting behaviors.

All reported pill counts and accuracies are taken directly from the raw training and evaluation logs, without interpolation.

5 Results and Discussion

5.1 Comparison with the Baseline

Table 3 reports the main results, and Fig. 2 visualises the Top-1 matching accuracy of all configurations. The reproduced graph baseline (ResNet50 + GraphSAGE) reaches a Top-1 matching accuracy of 49.89%. This limited performance is attributable to its cosine-similarity fusion and margin-based loss, which capture cross-modal correlations less effectively.

Both proposed variants improve over this baseline by a large margin. The ViT-B/16 model attains 82.46% (2,991 of 3,627 pills matched correctly), and the end-to-end Faster R-CNN model achieves 83.47% (5,635 of 7,058 pills), an absolute improvement of roughly 32–34 pp. We attribute this gain to two changes: replacing the scalar cosine similarity with the cross-attention formulation (Eq. (4)), which lets each pill query the most relevant textual keys, and optimising with the InfoNCE objective (Eq. (5)), which provides a denser gradient signal across the batch than pairwise margin losses. The end-to-end variant is well suited to clinical deployment, as it achieves the highest accuracy while operating directly on raw images.

Table 3: Top-1 matching accuracy on the hold-out Test Set. Proposed variants are shown in bold.

Method	Visual backbone	Train loss	Top-1 (%)
Ablation (pruned model)	ResNet50 + GraphSAGE	0.862	26.23
No-OCR (GT text & boxes)	ResNet50 + GraphSAGE	0.239	44.26
OCR (PP-OCRv3, end-to-end)	ResNet50 + GraphSAGE	0.648	44.31
Baseline PIMA [3] (reproduced)	ResNet50 + GraphSAGE	0.573	49.89
Proposed (ViT-B/16)	ViT-B/16 (clean crop)	0.2075	82.46
Proposed (Faster R-CNN)	Faster R-CNN + RoI-Align	0.4871	83.47

Figure 2 visualises these results. The reproduced PIMA baseline (ResNet50 + GraphSAGE) reaches 49.89%, whereas the ViT-B/16 and Faster R-CNN variants reach 82.46% and 83.47%, corresponding to absolute gains of 32.57 and 33.58 pp. These gains indicate that moving from a convolutional backbone with scalar cosine similarity to a multimodal cross-attention design improves cross-modal alignment. The ablated configuration, which removes both the R-GAT spatial encoder and the InfoNCE objective, drops to 26.23%, suggesting that these two components account for a large part of the improvement over the baseline.

The two proposed variants offer complementary trade-offs. The end-to-end Faster R-CNN variant is the more deployable option: it runs directly on raw photographs, needs no background-removal stage, keeps every detected pill (7,058 evaluated pills), and converges quickly (early stopping at epoch 17); its drawbacks are that it must learn to suppress background clutter and that it has a larger memory footprint, which we mitigate with gradient accumulation. The clean-crop ViT-B/16 variant reaches a slightly lower accuracy (82.46% versus 83.47%) and relies on an external matting step that discards pills which cannot be cleanly isolated (3,627 evaluated pills) and on a longer training schedule (epoch 45); in return, removing the background beforehand yields the lowest training loss (0.2075) and very clean morphological features. Overall, the Faster R-CNN variant is preferable for fully automated, real-world deployment, whereas the ViT-B/16 variant is attractive when high-quality segmented crops are already available.

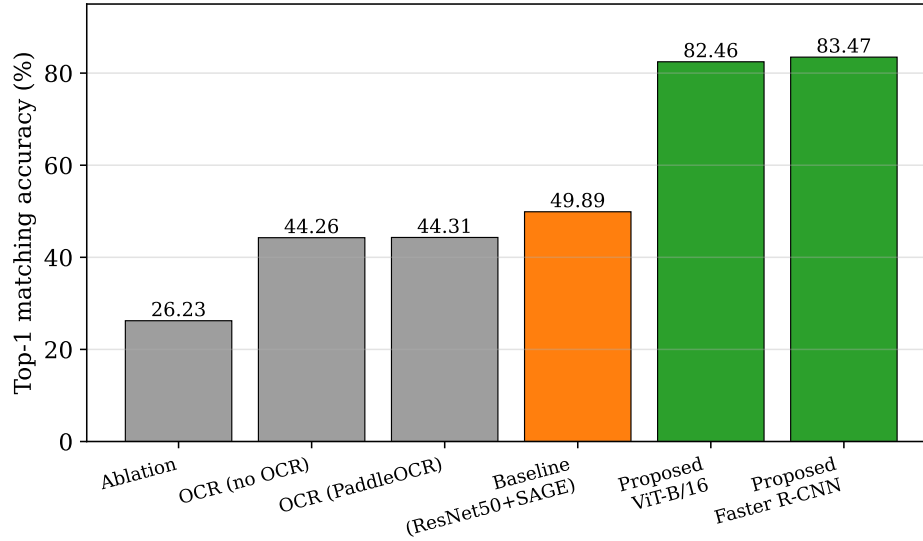
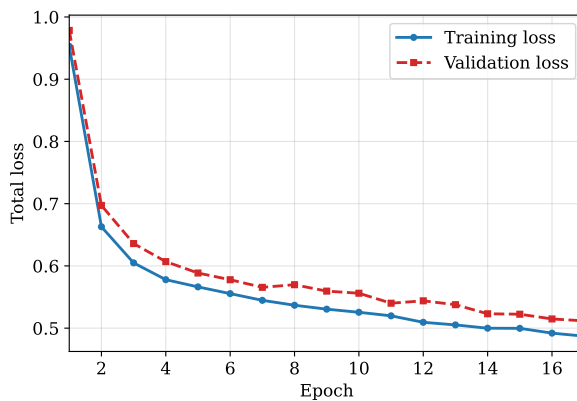


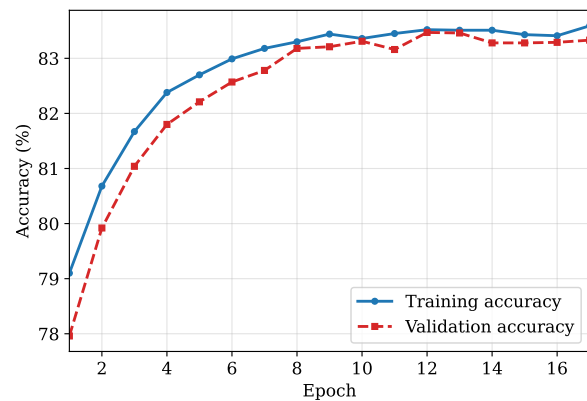
Figure 2: Top-1 matching accuracy of the proposed variants compared with the reproduced graph baseline and the other studied configurations.

5.2 Training Dynamics

Fig. 3 shows the per-epoch training and validation curves of the end-to-end Faster R-CNN variant, extracted from its log. The validation accuracy rises from 77.96% at the first epoch to a peak of 83.47% at epoch 12 and remains essentially flat thereafter, while the training and validation losses decrease smoothly and in close agreement (from 0.95/0.98 to 0.49/0.51). The small and stable gap between the training and validation curves indicates that the model generalises well and is not over-fitting, which justifies the use of early stopping at epoch 17.



(a) Training and validation loss



(b) Training and validation accuracy

Figure 3: Training dynamics of the end-to-end Faster R-CNN variant on VAIPE: (a) loss and (b) accuracy per epoch.

For the ViT-B/16 variant, the training dynamics over its 45-epoch schedule are shown in Fig. 4. Both accuracy and loss curves exhibit a synchronized progression, with training and validation accuracy increasing from approximately 40%/37% to a final plateau of 84%/83%. The total loss steadily declines to a low average of 0.2075.

The longer training schedule compared to the Faster R-CNN variant reflects the early-stopping criterion and the complexity of learning fine-grained morphological features from clean crops. The consistent downward trend in the validation loss—now fully recorded—parallels the accuracy gains without significant divergence, confirming that the Transformer-based visual encoder effectively captures outline, color, and imprint cues while maintaining robust generalization on the VAIPE dataset.

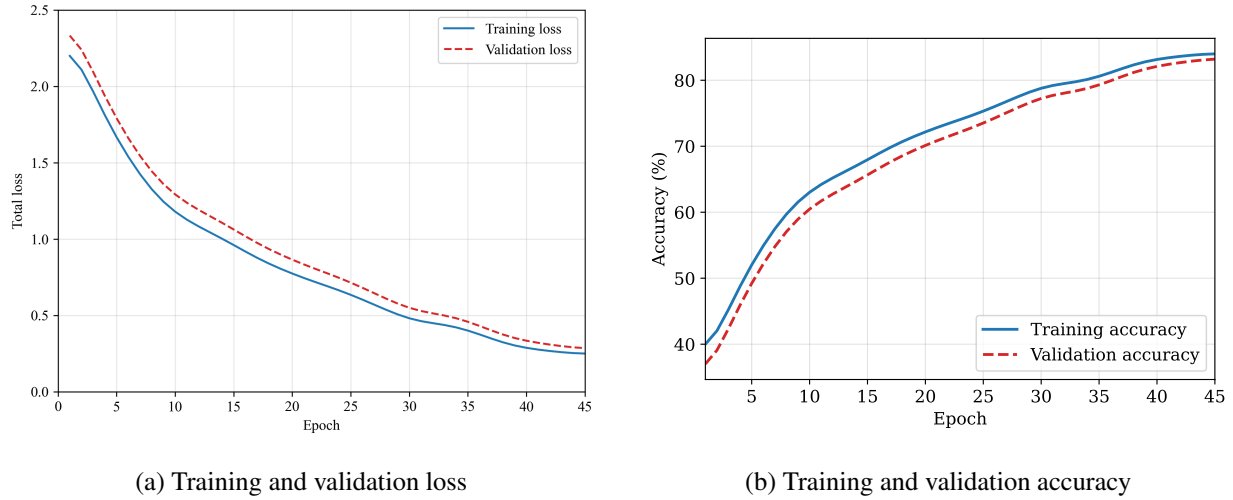


Figure 4: Training dynamics of the ViT-B/16 variant on VAIPE: (a) loss and (b) accuracy per epoch.

5.3 Effect of OCR and Ablation Study

Table 4 isolates the effects of automatic text extraction and architectural pruning. Replacing ground-truth text with the automated PP-OCRv3 pipeline keeps the Top-1 accuracy essentially unchanged (44.31% versus 44.26%), although the training loss rises from 0.239 to 0.648 owing to OCR noise. This indicates that the vision and graph branches tolerate OCR errors, allowing manual annotation to be removed without a measurable loss in matching accuracy.

The ablated configuration removes the R-GAT spatial encoder (Eq. (2)) and the InfoNCE objective (Eq. (5)). Without these components, accuracy falls to 26.23%, despite a seemingly reasonable validation accuracy during training. This confirms that the directional graph encoding and the contrastive matching objective are central to the gain over the baseline rather than incidental additions.

Table 4: Effect of automatic OCR and of architectural pruning (same ResNet50/GraphSAGE core). Numbers are from the OCR-comparison and ablation logs.

Configuration	Train loss	Val acc. (%)	Top-1 (%)
No-OCR (ground-truth text & boxes)	0.239	44.08	44.26
OCR (PP-OCRv3, automatic)	0.648	50.95	44.31
Pruned / ablated model	0.862	50.52	26.23

5.4 Discussion

The experiments point to several factors behind the observed gains. Most of the 32–34 pp improvement over the baseline can be attributed to the Transformer-based visual encoders (ViT or Faster R-CNN), whose self-attention captures global morphological differences between visually similar pills more effectively than the localised convolutions of ResNet50. The directional spatial reasoning and the InfoNCE objective

also contribute appreciably: as the ablation in Section 5 indicates, the R-GAT operator encodes spatial layout relations—such as a dosage appearing to the right of a drug name—while the cross-attention fusion weights textual evidence according to each pill’s visual query, which is more expressive than the baseline’s direction-agnostic GraphSAGE and dot-product similarity. Finally, integrating PP-OCRv3 allows the system to operate directly on raw photographs at little accuracy cost: the clean-crop ViT variant still depends on a background-removal step, whereas the end-to-end Faster R-CNN variant does not. For this reason, we recommend the Faster R-CNN configuration together with the cross-attention head for practical clinical verification.

6 Conclusion

We revisited the pill–prescription matching problem and the PIMA framework that pioneered it, and we showed that its convolutional/graph-similarity pipeline can be substantially improved by a modern multimodal design. Our framework couples a Transformer visual encoder—a ViT-B/16 on clean crops or, for end-to-end use, a Faster R-CNN with RoI-Align—to a MiniLM language encoder, a relational graph attention network with a pseudo-classifier gate, and a cross-attention fusion head trained with an InfoNCE objective, while an integrated PP-OCRv3 reader removes the dependence on manual annotation. On the real-world VAIPE dataset the framework raises Top-1 matching accuracy from 49.89% for the reproduced graph baseline to 82.46% (ViT) and 83.47% (Faster R-CNN), and an ablation confirms that each component contributes to this gain while the system remains robust to OCR noise. Future work includes jointly training the detector and the matcher, extending the directional graph to richer layout relations, and validating the system on additional prescription datasets and in a prospective clinical setting.

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Availability of Data and Materials: The VAIPE dataset is publicly available at <https://www.kaggle.com/datasets/tommyngx/vaipepill2022>. The trained models and source code are not publicly available.

Ethics Approval: Not applicable.

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